

Tutorial 1: CIA in Practice

Examining the following four scenarios, determine which principles of the CIA triad is or are directly threatened and explain why.

Case1 Prompt Poaching

In early 2026, cybersecurity researchers have discovered two new malicious extensions on the Chrome Web Store that are designed to exfiltrate OpenAI ChatGPT and DeepSeek conversations alongside browsing data to servers under the attackers' control. The names of the extensions, which collectively have over 900,000 users, are below:

- Chat GPT for Chrome with GPT-5, Claude Sonnet & DeepSeek AI (600,000 users)
- AI Sidebar with Deepseek, ChatGPT, Claude, and more. (300,000 users)

The findings follow weeks after Urban VPN Proxy, another extension with millions of installations on Google Chrome and Microsoft Edge, was caught spying on users' chats with artificial intelligence (AI) chatbots. This tactic of using browser extensions to stealthily capture AI conversations has been codenamed Prompt Poaching by Secure Annex.

Case 2: DDoS Attack

France's national postal service (La Poste) and its banking subsidiary (La Banque Postale) were knocked offline on 22 December 2025 due to a suspected distributed denial-of-service (DDoS) attack. The postal service called the attack "a major network incident" that was disrupting "all of our information systems."

La Poste's online mail and banking services, website, and mobile app are among the services that are temporarily unavailable, per a machine translation of the announcement, which said customers could still carry out banking and postal transactions in person.

The banking branch of the company La Banque Postale also published an announcement about the cyberattack, which was "temporarily making our customers' access to their mobile app and online banking space unavailable."

Case 3: Ransomware

Ransomware is a form of malicious cyberattack that involves encrypting a victim's data or locking them out of their systems, with the attacker demanding a ransom payment to restore access. Historically, these attacks focused primarily on encrypting files to disrupt business operations. However, they have increasingly evolved into "double extortion" schemes. In this more dangerous variant, attackers not only encrypt the victim's data to stop them from using it but also steal sensitive information. They then threaten to release or sell this stolen data on the dark web if the ransom is not paid, significantly raising the stakes for the victim.

Case 4: Data Poisoning

Data poisoning is a sophisticated and insidious attack strategy in which adversaries subtly inject malicious samples, incorrect labels, or hidden backdoors into the datasets used to train artificial intelligence models. By tampering with the training data, attackers can corrupt the model's logic, causing it to learn wrong patterns or make flawed decisions. For example, a poisoned spam filter might be manipulated to classify emails containing specific keywords like "urgent transfer" as safe, thereby allowing malicious emails to bypass security filters and facilitating financial fraud. This undermines the reliability of the AI system without the users realizing the decision-making process has been altered.